

Excellent! **Node Affinity** is an advanced version of **Node Selector**. If you understand Node Selector first, Node Affinity becomes very easy.

What is Node Affinity?

Simple Definition

Node Affinity is a Kubernetes feature that allows you to specify rules or preferences for scheduling Pods on particular Nodes based on Node labels.

Easy Memory

 **Node Affinity = Smart Node Selector**

It uses **Node Labels** just like Node Selector, but it gives you **more control and flexibility**.

Real-Life Example

Imagine a school with three classrooms.

Classroom A → Science

Classroom B → Computer

Classroom C → English

A Computer student says:

"I want to study in the Computer classroom."

Kubernetes checks the classroom labels.

If it matches, the student goes there.

This is **Node Affinity**.

Why Do We Need Node Affinity?

Suppose your cluster has three Nodes.

Node 1

Label:

disk=ssd

Node 2

Label:

disk=hdd

Node 3

Label:

gpu=true

Some applications need:

- SSD storage
- GPU
- High Memory

Node Affinity helps Kubernetes schedule Pods on Nodes that have the required labels.

Node Selector vs Node Affinity

Node Selector

nodeSelector:

disk: ssd

Very simple.

Only exact matching.

Node Affinity

affinity:

nodeAffinity:

More powerful.

Supports:

- Hard rules
 - Soft preferences
 - Multiple conditions
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Types of Node Affinity

There are **2 main types**.

Type 1

RequiredDuringSchedulingIgnoredDuringExecution

Simple Definition

This is a Hard Requirement. The Pod will only run on a Node that matches the specified labels.

If no matching Node exists,

 the Pod will **NOT** run.

It stays in **Pending** state.

Real-Life Example

Imagine a student says:

"I will **ONLY** study in the Computer classroom."

If the Computer classroom is full,

The student waits.

He does **not** go to another classroom.

This is a **Hard Requirement**.

Kubernetes Example

Nodes

Node 1

disk=ssd

Node 2

disk=hdd

Pod requires

disk=ssd

Scheduler checks.

Node 1

 Matches

Pod runs.

Suppose only

Node 2

disk=hdd

exists.

Pod requires

disk=ssd

Result

 Pod stays Pending.

Easy Memory

 **Required = MUST Match**

Type 2

PreferredDuringSchedulingIgnoredDuringExecution

Simple Definition

This is a **Soft Requirement**. Kubernetes tries to schedule the Pod on a matching Node, but if no matching Node exists, it can choose another Node.

Real-Life Example

A student says:

"I prefer the Computer classroom."

If it's available,

Go there.

If it's full,

Go to another classroom.

This is a **Soft Requirement**.

Example

Nodes

Node 1

disk=ssd

Node 2

disk=hdd

Pod prefers

disk=ssd

If Node 1 is available



If Node 1 is unavailable

Easy Memory

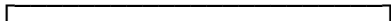


Visual Diagram

Scheduler



Node Affinity Rules



Required (Hard)

Preferred (Soft)

MUST Match

Try to Match



No Match 

No Match

Pending

Run on Another Node

YAML Example

Required (Hard)

affinity:

nodeAffinity:

requiredDuringSchedulingIgnoredDuringExecution:

Meaning:

👉 The Pod **must** run on a matching Node.

Preferred (Soft)

affinity:

nodeAffinity:

preferredDuringSchedulingIgnoredDuringExecution:

Meaning:

👉 The Pod **prefers** a matching Node, but another Node is acceptable if needed.

What Does "Ignored During Execution" Mean?

This part confuses many beginners.

Let's break it down:

- **During Scheduling** = When Kubernetes is deciding where to place the Pod.
- **During Execution** = After the Pod is already running.

"Ignored During Execution" means:

Kubernetes checks the rule **only when scheduling the Pod**. If the Node's label changes later, Kubernetes does **not** move the running Pod.

Example

1. Pod is scheduled on a Node with disk=ssd. 
2. Later, someone changes the label to disk=hdd.

3. The Pod **keeps running** on that Node.

Kubernetes ignores the label change after the Pod has already started.

Node Selector vs Node Affinity

Node Selector	Node Affinity
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Simple	Advanced
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Exact label match	Flexible matching rules
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One type	Hard & Soft rules
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Easy to use	More powerful
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Required vs Preferred

Required (Hard)	Preferred (Soft)
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Must match	Prefer to match
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No matching Node → Pod stays Pending	No matching Node → Pod can run elsewhere
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Strict rule	Flexible rule
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Quick Revision Notes

Node Affinity

- Uses **Node Labels**.
- Controls where Pods are scheduled.
- More flexible than Node Selector.

Types

RequiredDuringSchedulingIgnoredDuringExecution

- Hard Requirement.
- Pod **must** find a matching Node.

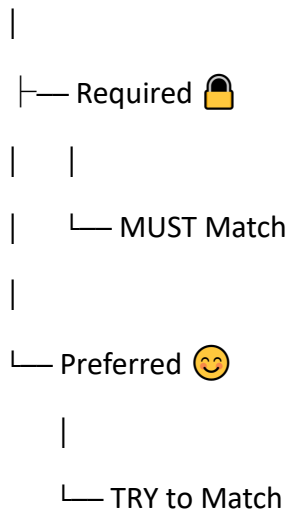
- Otherwise, it stays **Pending**.

PreferredDuringSchedulingIgnoredDuringExecution

- Soft Requirement.
- Kubernetes **tries** to find a matching Node.
- If none is available, it schedules the Pod on another suitable Node.

Easy Memory Trick ★

Node Affinity



One-Line Memory Trick

- 🔒 Required = Must run on a matching Node.
- 😊 Preferred = Try to run on a matching Node, but another Node is acceptable if necessary.